FlowFigTabMiner: Multimodal Extraction of Structured Flow Chemistry Data from Figures, Tables, and Text Enables Organolithium Lifetime Prediction

Wenyuan Zhao, Xianzhu Zhong, Yosuke ASHIKARI, Hiromichi V. MIYAGISHI, Mohmmad S. QENAWY, Simeng Wang, Jouffery Xiaer, Yirui Peng, and Aiichiro Nagaki*

Department of Chemistry, Faculty of Science, Hokkaido University, Sapporo 060-0810, Japan

E-mail: nagaki@sci.hokudai.ac.jp

Abstract

Quantitative reaction data in flow chemistry literature is predominantly stored in figures (yield-vs-residence-time heatmaps) and tables (with embedded molecular structure images), formats inaccessible to conventional text mining. We present FlowFigTabMiner, a fully automated multimodal pipeline that chains Florence-2 for figure/table detection, four custom YOLO models for element segmentation, PaddleOCR for text recognition, MolNexTR for image-to-SMILES conversion, and LLM adjudication for evidence integration. On a 10-figure/10-table benchmark, the pipeline achieves an overall extraction F1 of 0.838, outperforming Gemini 3.1 Pro (0.833) and Claude Sonnet 4.6 (0.806), with a dominant advantage on figure extraction (F1 = 0.892

vs 0.575/0.561). Applied to 125 organolithium flow chemistry publications, the pipeline constructs a 1,470-row kinetics dataset—the first with flow-condition coverage (residence time, temperature, yield). Kinetic modeling of the extracted yield-vs-t_R data yields decomposition activation energies (E_a) for 14 organolithium intermediates spanning nearly four orders of magnitude in half-life. A unified four-parameter model ($E_a = -48.0\sigma + 8.0\,E_s + 25.5\,\delta_{\text{ortho}} + 44.4\,\delta_{\text{benzyne}} + 35.9$ kJ/mol, $R^2 = 0.980$, $n = 10$) integrating Hammett electronic, Taft steric, Li$\cdots$O chelation, and benzyne-elimination effects enables quantitative half-life prediction for all 10 ArLi intermediates from substituent constants alone. Arrhenius extrapolation to batch conditions (−78 °C, 600 s) predicts intermediate survival within 3% of experimental yields, validating the extracted kinetic parameters for reactor selection.

INTRODUCTION

Recent advances in generative artificial intelligence, particularly large language models (LLMs), are expanding the applications of machine learning in chemical engineering.[1–5] These models have demonstrated significant potential across diverse tasks, such as assisting with process flow diagrams,[6] predicting thermochemical and kinetic properties,[7] aiding process development,[8] and optimizing reaction conditions or yields.[9–12] However, a primary bottleneck limits the broader application of these AI models: the lack of high-quality, large-scale training datasets.[7,13,14] In response to this challenge, significant efforts are underway to improve data standardization and accessibility. For example, Kearnes et al. have pioneered the development of reaction databases and defined standardized formats to promote effective data sharing and reuse.[16] Kang et al. successfully addressed the data scarcity challenge by employing a chain of advanced LLMs to mine over 40,000 articles, constructing a comprehensive, structured dataset for metal-organic frameworks (MOFs).[17]

A critical gap persists for flow chemistry data in particular. Advancements in laboratory automation, including high-throughput experimentation (HTE)[18,19] and flow chemistry,[20–22] can produce standardized experimental data at scale. However, the published flow chemistry literature, while rich in information, stores critical reaction parameters not only in text but also in *figures* (yield-vs-residence-time curves, temperature–time heatmaps) and *tables* (often containing molecular structure images rather than text-based SMILES). Existing structured databases such as USPTO and the Open Reaction Database (ORD)[15] lack flow-specific parameters—residence time, flow rate, and reactor geometry—that are essential for process optimization.[23] As a result, publicly available, structured databases for flow chemistry remain remarkably scarce, directly hindering the development of robust machine learning models tailored for continuous manufacturing.

Over the past few decades, various methods have been developed to extract structured information from unstructured chemical literature. Leveraging natural language processing (NLP) techniques to build systems that can automatically and effectively process lengthy chemical texts, including scientific literature or patents, in order to extract key information has received widespread attention.[24] Supervised learning models like Google's BERT language model[25] and its domain-specific variants such as SciBERT,[26] ChemBERTa,[27] BioBERT,[28] and MatSciBERT[29] have been successfully applied to various chemical text mining tasks.

For instance, OPSIN[30] serves as an early exemplar, utilizing a heuristic-based method that underscores the potential benefits of applying NLP to chemical data extraction. ChemDataExtractor[31] pioneered an automated approach to extract chemical information from scientific literature by integrating NLP and machine learning techniques. Subsequently, ChemRxnBERT[32] harnessed the power of pre-training on chemical literature to foster a deeper understanding of chemical context. More recently, OpenChemIE[33] introduced an open-source toolkit that automatically extracts chemical reaction data by integrating information from text, tables, and figures. Xiang et al.[34] presented a workflow that employs a local LLMs combined with multi-step prompt engineering to automatically extract synthesis parameters for on-surface reactions. However, it does not capture process-level information such as reactor types or influencing process parameters. Pan et al.[35] introduced Chat-microreactor, an LLM-based assistant that extracts fluidic properties and microchannel dimensions from literature to train machine learning models for classifying multiphase flow patterns (e.g., dripping, jetting). Nevertheless, this tool is explicitly prompted to ignore chemical reaction data and therefore does not extract critical process development parameters such as catalyst, temperature, or yield. Critically, all of these tools operate exclusively on text, leaving the quantitative data encoded in scientific figures and molecular structures embedded in tables entirely untapped.

Vision-Language Models (VLMs) such as GPT-4V, Gemini, and Claude have emerged as promising multimodal extractors capable of "seeing" figures and tables directly.[36,37] However, our evaluation reveals that these models hallucinate coordinates when reading dense scatter plots, with F1 scores dropping below 0.36 on complex heatmaps and multi-series figures (Section). VLMs approximate rather than precisely read pixel positions, leading to systematic errors in quantitative data extraction. Furthermore, commercial safety guardrails frequently block legitimate flow chemistry queries involving high-energy reactions (e.g., nitration, azides) or hazardous conditions, severely limiting their utility for process development literature.[38] No existing end-to-end system addresses the full pipeline from PDF ingestion through figure/table detection, element segmentation, coordinate extraction, and molecular structure recognition to a structured database.

Herein, we introduce **FlowFigTabMiner**, a fully automated multimodal pipeline for extracting structured process data from flow chemistry literature. The framework chains Florence-2 for figure/table detection, four custom-trained YOLO models for element segmentation, the Table Transformer (TATR) for cell structure recognition, PaddleOCR for text extraction, MolNexTR for image-to-SMILES molecular structure conversion, and LLM-based adjudication for evidence merging. Three key advances are demonstrated: (1) YOLO-based coordinate mapping from scientific figures that surpasses state-of-the-art VLMs, achieving an extraction F1 of 0.892 compared to 0.575 for Gemini 3.1 Pro and 0.561 for Claude Sonnet 4.6;

(2) integrated table extraction with molecular structure recognition, enabling SMILES conversion from structure images; (3) construction of a 1,470-row organolithium kinetics dataset and a 1,267-row scope table dataset from 125 publications—the first with flow-condition coverage (residence time, temperature, yield). We further demonstrate the dataset's chemical utility by extracting decomposition activation energies (E_a) for 14 organolithium intermediates and establishing quantitative Hammett (E_a vs σ, $r = -0.979$) and Taft ($t_{1/2}$ vs E_s, $r = -0.969$) correlations that predict ArLi intermediate stability from substituent constants. The kinetic parameters are validated by Arrhenius extrapolation to independent batch scope data, confirming quantitative predictive accuracy. The pipeline, models, dataset, and a cloud-deployed web tool are open-sourced.

RESULTS AND DISCUSSION

Pipeline Architecture

FlowFigTabMiner is an end-to-end pipeline that converts flow chemistry PDF publications into structured reaction databases (Figure 1). The system ingests a PDF document and processes it through three parallel extraction tracks—figure, table, and text—whose outputs are merged by an LLM adjudicator into normalized JSON records, followed by manual inspection before the final dataset is released for downstream machine learning and validation.

Input layer: PDF parsing and region detection. Each PDF page is analyzed by TF-ID (Florence-2-base fine-tuned for scientific document layout analysis), which detects and crops all figures and tables as individual images. Simultaneously, PyMuPDF extracts the full text of the document, preserving section structure and captions. The cropped images are routed to domain-specific processing tracks: FigureDataMiner for scatter plots, line charts, and heatmaps (Figure 2), and TableDataMiner for structured reaction tables containing numerical data and molecular structure images (Figure 3).

Global and local parameters pool. A key design feature is the hierarchical parameters

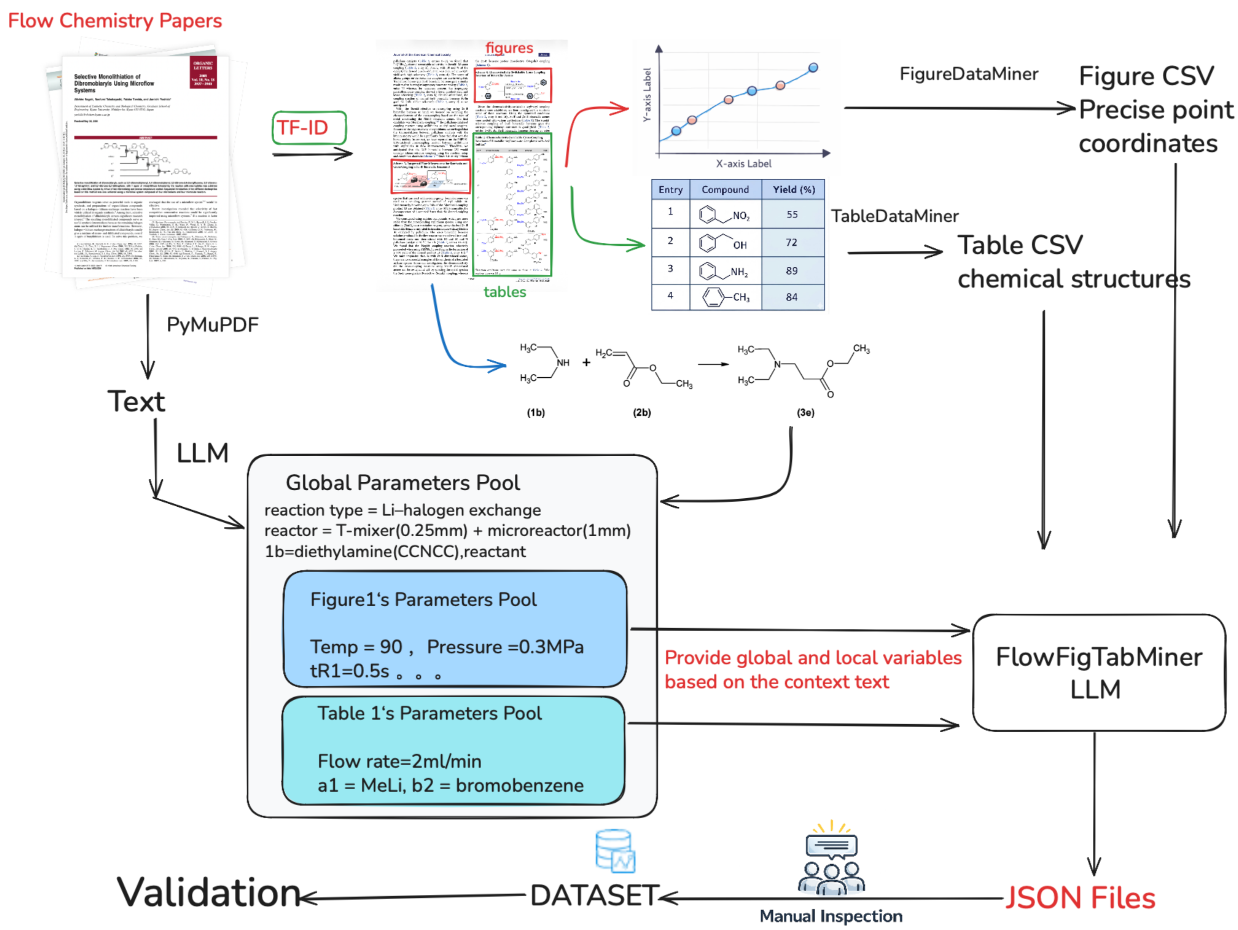


Figure 1: Overview of the FlowFigTabMiner pipeline. Flow chemistry papers are parsed by TF-ID to detect figures and tables, while PyMuPDF extracts the full text. FigureDataMiner and TableDataMiner produce structured CSVs with precise coordinates and chemical structures, respectively. An LLM constructs a hierarchical parameters pool (global and local) from the text context, and merges all evidence into JSON records. Manual inspection ensures data quality before final dataset release.

pool that captures contextual information from the surrounding text. The LLM first extracts *global parameters* that apply to all experiments in a paper—such as reaction type (e.g., Li–halogen exchange), reactor specification (e.g., T-mixer 0.25 mm + microreactor 1 mm ID), and general conditions. It then extracts *local parameters* specific to each figure or table, including temperature, pressure, residence time, flow rate, and compound identifiers (e.g., "a1 = MeLi, b2 = bromobenzene"). This two-level architecture ensures that shared conditions stated once in the experimental section are propagated to all associated data points, while figure- or table-specific variables override the global defaults.

Figure processing track. Figure crops undergo four sequential steps (Figure 2). First, a YOLO macro segmentation model (v11m-fig-seg, 6 classes: caption, legend, subfigure marker, target image, x-axis title, y-axis title) identifies chart elements and removes non-chart regions. Second, a YOLO micro detection model (v11m-fig-sca, 4 classes: data point, data value, x-tick label, y-tick label) locates all quantitative elements within the cleaned chart. Third, a coordinate mapping algorithm applies OCR to tick labels, constructs pixel-to-physical axis transforms via RANSAC regression, and maps each detected data point to physical coordinates $(X, Y_{\text{left}}, Y_{\text{right}})$. Log scales are detected automatically when >50% of tick labels match scientific notation patterns (e.g., 10^{-2}). Fourth, legend matching assigns series labels to data points by color and spatial proximity. The pipeline handles scatter plots, line charts, bar charts, and yield-vs-t_R heatmaps, the last detected by a legend text pattern (`\d+~\d+%`).

Table processing track. Table crops are processed in six steps (Figure 3). A YOLO table segmentation model (v11m-tab-seg, 4 classes: table body, caption, note, reaction scheme) isolates the table body. A second YOLO model (v11s-tab-mol, 1 class: molecular structure) detects embedded structure images, which are converted to SMILES by MolNexTR and then masked with white fill to prevent OCR interference. The Table Transformer (TATR, microsoft/table-transformer-structure-recognition-v1.1-all) detects rows, columns, and headers, generating a cell grid. Each cell is classified as text (OCR via PaddleOCR PP-OCRv4 rec-only) or molecule (SMILES from the masking step, matched by IoU > 0.5). The assembled grid is exported as CSV with keyword-based relevance filtering. In parallel, the reaction scheme regions isolated by tab-seg are processed by a dedicated scheme segmentation model (v11n-tab-scheme-seg, 4 classes: arrow, molecule, table-condition, table-mark). This model detects molecular structures, reaction arrows, condition text, and compound labels (e.g., “2a”, “3b”) within scheme diagrams. Detected molecules are converted to SMILES via MolNexTR, and each is assigned a role—reactant or product—based on its spatial position relative to the arrow. The resulting *compound pool* maps compound labels to SMILES strings (e.g., {"2a": "BrC1=CC=CC=C1"}), serving as a lookup table that enables the LLM adjudicator to resolve abbreviated compound references in the table body.

LLM adjudication. The FlowFigTabMiner LLM (Qwen-3.5-Plus, temperature = 0.1) receives figure evidence (CSV of coordinates and series labels), table evidence (CSV of cell contents and SMILES), the global and local parameters pools, and the extracted PDF text. It merges these heterogeneous inputs into structured reaction records with standardized fields (yield, temperature in °C, residence time in seconds, flow rate in mL/min, SMILES). Unit normalization, SMILES validation via RDKit, and cross-source deduplication are applied automatically. The output is a set of JSON files, one per paper, containing all extracted reaction entries.

Manual inspection and dataset release. The JSON outputs undergo manual inspection by domain experts, who verify extracted values against the original publications, correct systematic errors (e.g., misassigned compound identifiers or unit conversion mistakes), and flag ambiguous entries. This human-in-the-loop step is essential for ensuring data quality before the final dataset is released for machine learning applications. The pipeline is deployed as four microservices on Google Cloud Run (tfid, figure, table, frontend) with GCS-mounted model weights, providing a public web interface for end-to-end PDF-to-database extraction.

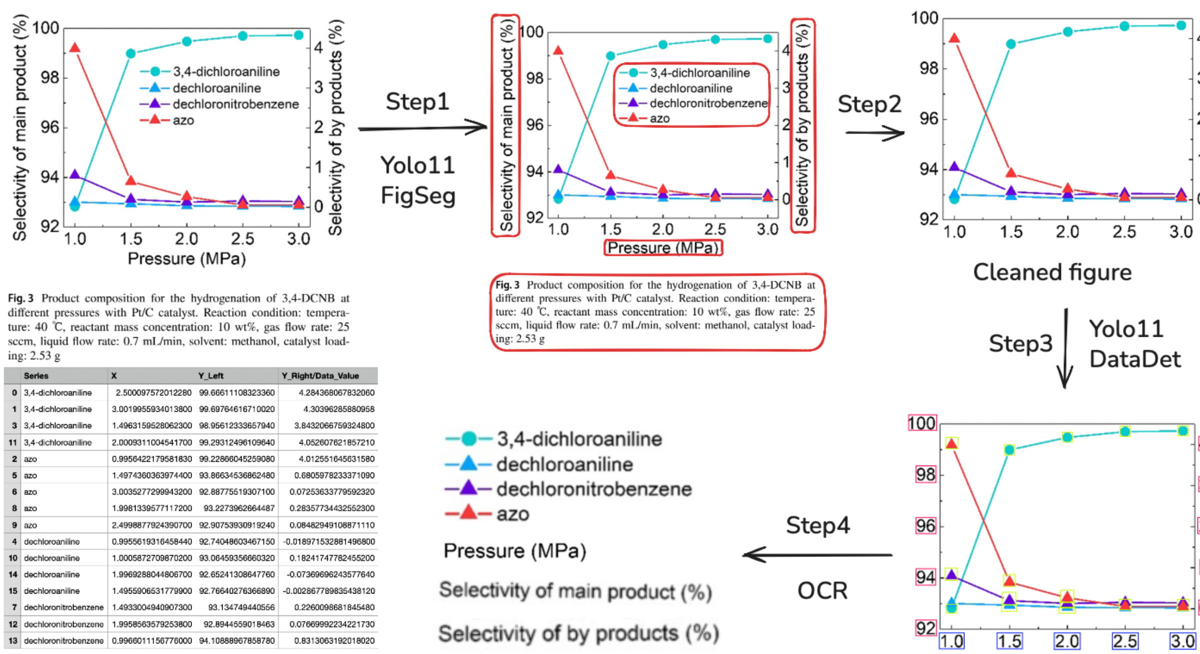
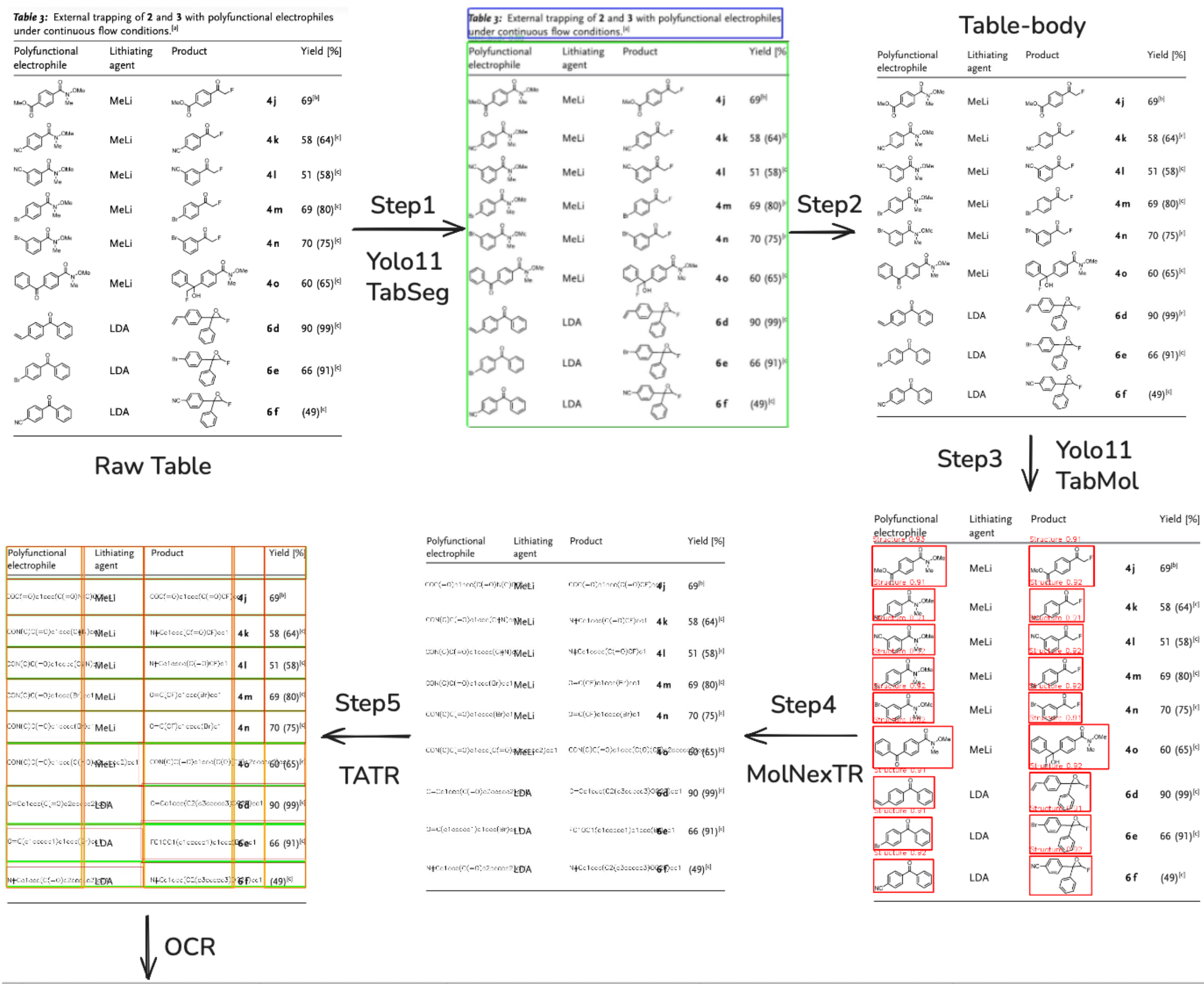


Figure 2: Overall architecture of the FlowFigTabMiner pipeline for figure data extraction. The workflow processes PDF documents through TF-ID detection, YOLO-based macro segmentation and micro detection, coordinate mapping, and legend matching to produce structured data points.



	0	1	2	3	4
0	Polyfunctional	Lithiating	Product		Yield [%]
1	COC(=O)c1ccc(C(=O)N(C)OC)cc1	MeLi	COC(=O)c1ccc(C(=O)CF)cc1	4j	69[b]
2	CON(C)C(=O)c1ccc(C#N)cc1	MeLi	N#Cc1ccc(C(=O)CF)cc1	4k	58 (64)[c]
3	CON(C)C(=O)c1cccc(C#N)c1	MeLi	N#Cc1cccc(C(=O)CF)c1	4l	51 (58)[c]
4	CON(C)C(=O)c1ccc(Br)cc1	MeLi	O=C(CF)c1ccc(Br)cc1	4m	69 (80)]
5	CON(C)C(=O)c1cccc(Br)c1	MeLi	O=C(CF)c1cccc(Br)c1	4n	70 (75)[c]
6	CON(C)C(=O)c1ccc(C(=O)c2ccccc2)cc1	MeLi	CON(C)C(=O)c1ccc(C(O)(CF)c2ccccc2)cc1	4o	60(65)[c]
7	C=Cc1ccc(C(=O)c2ccccc2)cc1	LDA	C=Cc1ccc(C2(c3ccccc3)OC2F)cc1	6d	90(99)[c]
8	O=C(c1ccccc1)c1ccc(Br)cc1	LDA	FC1OC1(c1ccccc1)c1ccc(Br)cc1	6e	66(91)[c]
9	N#Cc1ccc(C(=O)c2ccccc2)cc1	LDA	N#Cc1ccc(C2(c3ccccc3)OC2F)cc1	6f	(49)[c]

Figure 3: Table extraction sub-pipeline. YOLO detects table regions and molecular structures; structures are masked before TATR cell detection; PaddleOCR reads text cells while MolNexTR converts structure images to SMILES.

YOLO Model Training and Object Detection

Five YOLO models were trained on the YOLOv11 architecture (Ultralytics)[39] using manually annotated chemical figures and tables from the flow chemistry literature. Training images were annotated with bounding boxes by domain experts, and all models used 1024×1024 px input resolution. Three models employed the medium backbone (v11m) for region-level segmentation tasks, while smaller backbones (v11s and v11n) were used for tasks with fewer classes.

The five models address complementary detection tasks across the figure and table tracks. **v11m-fig-seg** performs macro segmentation of figure images into 6 classes (caption, legend, subfigure marker, target image, x-axis title, y-axis title), isolating the data-bearing region from surrounding annotations. **v11m-fig-sca** detects 4 micro-level classes (data point, data value, x-tick label, y-tick label) required for coordinate reconstruction. **v11m-tab-seg** segments table images into 4 regions (table body, caption, footnote, reaction scheme), enabling separate processing of structured data and molecular diagrams. **v11s-tab-mol** detects a single class (molecular structure) within table body regions, feeding crops to MolNexTR for SMILES conversion. **v11n-tab-scheme-seg** parses reaction scheme diagrams into 4 classes (arrow, molecule, table-condition, table-mark), enabling automatic construction of compound pools that map abbreviated labels to molecular structures.

All five models achieve mAP_{50} above 91.7% (Table 1). The tab-mol model attains the highest precision (96.2%) and mAP_{50} (95.6%), reflecting the visually distinctive features of chemical structures against table backgrounds. The fig-sca model achieves 92.5% mAP_{50}, which is critical because coordinate mapping accuracy depends directly on reliable tick-label and data-point localization. The fig-seg model shows the lowest $\text{mAP}_{50\text{–}95}$ (66.9%), attributable to the high variability in axis title positioning and font size across different journals and figure styles.

Table 1: Comparative experimental results of different YOLO models for FlowFigTabMiner. The best performance is indicated in bold.

Models	Precision/%	Recall/%	F1 Score/%	mAP50/%	mAP50–95/%
v11m-fig-seg	88.2	91.7	89.9	91.7	66.9
v11m-fig-sca	92.1	91.9	92.0	92.5	69.3
v11m-tab-seg	91.2	89.9	90.5	94.8	77.7
v11s-tab-mol	**96.2**	**95.4**	**95.8**	**95.6**	**81.2**
v11n-tab-scheme-seg	93.9	94.3	94.1	96.5	72.1

Three-Way Benchmark: Pipeline vs. State-of-the-Art VLMs

To contextualize FlowFigTabMiner's performance, we conducted a three-way comparison against two state-of-the-art vision–language models: Gemini 3.1 Pro and Claude Sonnet 4.6. A test set of 10 figures (F1–F10) and 10 tables (T1–T10) was selected from diverse flow chemistry publications. Due to the cost of VLM evaluation, VLMs were benchmarked on the first five figures and five tables (F1–F5, T1–T5), while the pipeline was evaluated on all 20 samples. Ground-truth annotations were prepared independently by doctoral researchers. For figure data, predicted numerical values were matched to ground truth via greedy nearest-neighbor assignment with a 5% relative tolerance; for tables, exact string matching was used; SMILES correctness was assessed by RDKit canonical form validation.

On figure extraction (Table 2), the pipeline achieved F1 = 0.892 (precision 0.876, recall 0.932), outperforming Gemini (F1 = 0.575) and Claude (F1 = 0.561) by over 55%. VLM performance collapsed on heatmaps (F3, Gemini F1 = 0.36) and dense scatter plots (F5, Claude F1 = 0.19), where spatial precision is paramount. The root cause is fundamental: VLMs estimate coordinate values from visual context, while YOLO-based detection reads precise pixel positions that are then converted to physical units through calibrated axis transforms.

For table extraction, VLMs were clearly superior: Claude achieved F1 = 0.99 and Gemini 0.98, compared to 0.827 for the pipeline. The pipeline's table performance was limited by occasional TATR row/column misalignment on complex multi-header tables (T3, T4). Similarly, for SMILES extraction, Gemini (F1 = 0.944) and Claude (F1 = 0.867) outperformed the pipeline (F1 = 0.795). Notably, the tasks differ: the pipeline must convert molecular structure *images* to SMILES via MolNexTR (an image-to-sequence model), whereas VLMs read SMILES *text* already present in table cells.

Overall, the pipeline (F1 = 0.838) narrowly surpassed Gemini (0.833) and Claude (0.806), driven entirely by its dominant figure extraction performance. The key insight is that the strengths are complementary: the pipeline excels at extracting precise numerical data from figures, where pixel-level accuracy matters, while VLMs excel at structured text recognition from tables. From a practical standpoint, the pipeline runs fully locally without per-image API costs, making it suitable for large-scale corpus processing.

Table 2: Overall three-way comparison: FlowFigTabMiner vs. VLMs on 5 figures and 5 tables. Per-task breakdown (Figure, Table, SMILES) is provided in Table S5 of the Supporting Information.

Task	Metric	FlowFigTabMiner	Gemini 3.1 Pro	Claude Sonnet 4.6
Figure (F1–F5)	F1	**0.892**	0.575	0.561
Table (T1–T5)	F1	0.827	0.98	**0.99**
SMILES (T1–T5)	F1	0.795	**0.944**	0.867
Overall	Precision	**0.881**	0.818	0.799
	Recall	0.814	**0.861**	0.828
	F1	**0.838**	0.833	0.806

Organolithium Dataset Construction

Organolithium intermediates were selected as the target domain because their kinetic instability (half-lives ranging from milliseconds to minutes) makes precise residence-time control critical and flow chemistry the method of choice.[40] A corpus of 125 papers was assembled by keyword filtering ("organolithium", "aryllithium", "Br/Li exchange", "flow reactor", etc.; see Supporting Information for the complete keyword list), spanning publications from 2005 to 2024. Each paper was processed through the full FlowFigTabMiner pipeline (figure + table tracks), followed by LLM adjudication and human verification.

The resulting dataset comprises 3,195 reaction entries with field completeness as follows: molecular structure 15.1%, reaction outcome 61.3%, yield 49.3%, temperature 71.5%, residence time 56.7%, solvent 70.9%, and flow conditions 85.1%. After filtering for entries containing at least yield, temperature, or residence time, 2,884 modeling-ready rows remain, of which 1,688 include non-null residence times suitable for kinetic analysis. A unique feature is the inclusion of 249 heatmap-derived data points from 12 figure groups, where yield values were extracted from color-encoded scatter plots rather than tabulated text.

Figure 4 compares our dataset with two existing organolithium resources. The USPTO-LLM subset (6,073 rows from patent text) provides near-complete structural coverage but contains no reaction outcomes or flow-specific variables. The Open Reaction Database (ORD) organolithium subset (43,004 rows) offers 44.3% yield coverage but likewise lacks flow conditions entirely. Our dataset is the only one with substantial flow-condition coverage (85.1% flow, 4.5% batch) and the only one combining structure, outcome, and continuous process variables in a single resource, making it uniquely suited for data-driven flow process optimization.

Data quality was ensured through a human-in-the-loop verification workflow. SMILES enrichment was performed conservatively: only exact compound-name-to-SMILES matches from the ORD were accepted, with no algorithmic name-to-structure conversion. Over 15 organolithium reaction types (Br/Li exchange, halogen–metal exchange, directed metalation, carbolithiation, etc.) were classified to support reaction-type-specific modeling.

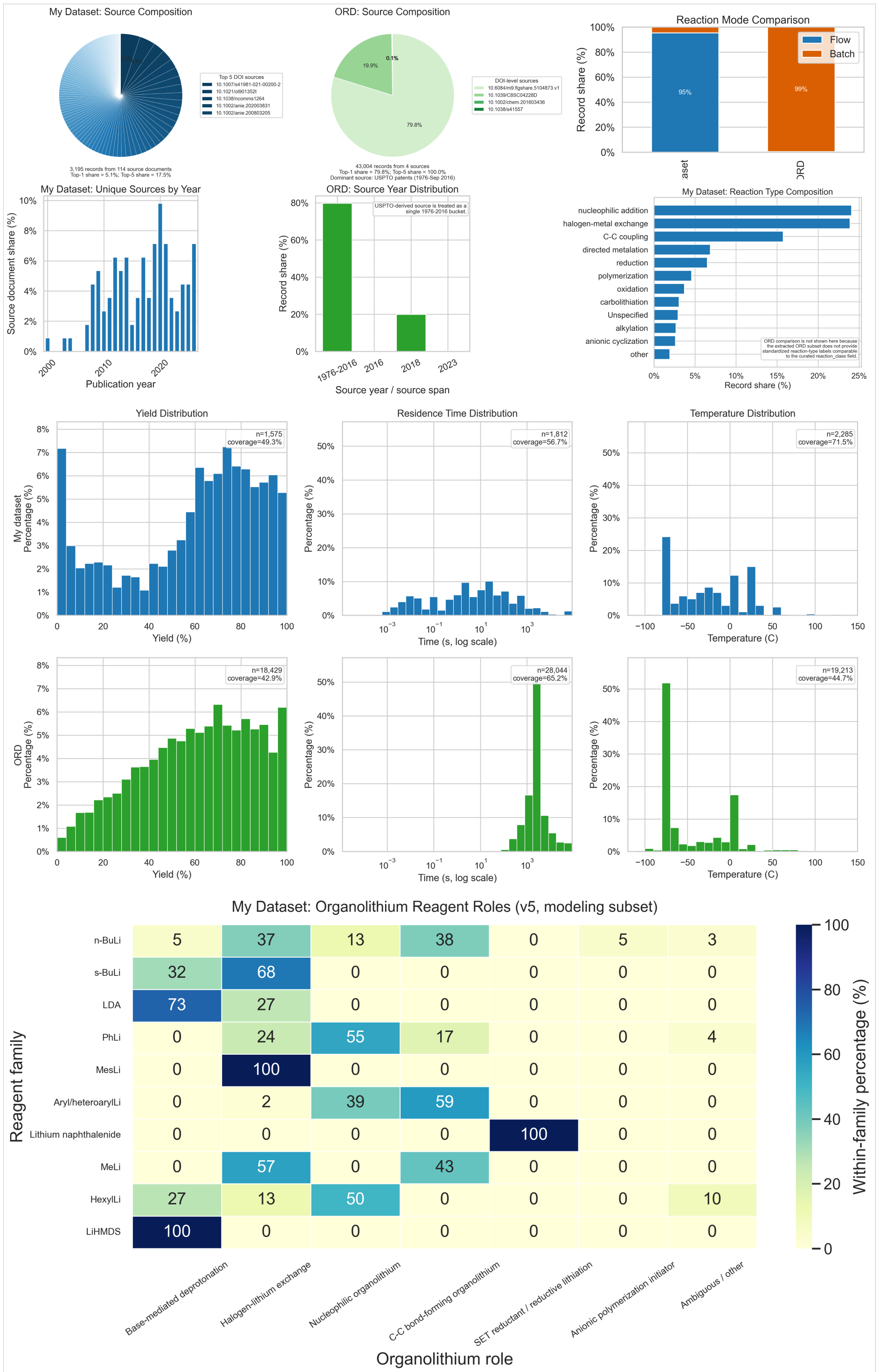


Figure 4: Organolithium dataset overview. (a) Field completeness comparison across our dataset, USPTO-LLM, and ORD. (b) Reaction type composition of the 3,195-row dataset.

Organolithium Intermediate Lifetime Prediction

Organolithium intermediates are kinetically unstable species whose lifetime governs the optimal residence time (t_R) and, consequently, the choice between batch, flow, or flash chemistry reactors.[40] Yield-vs-t_R heatmaps, ubiquitous in flow chemistry publications, encode this kinetic information in their shape: a rising phase (formation-limited), a peak, and a decay phase (decomposition-limited). Extracting quantitative kinetics from these figures is precisely the task for which FlowFigTabMiner was designed (figure F1 = 0.892).

Half-life extraction from yield decay curves

From 20 papers containing yield-vs-t_R heatmaps, the pipeline extracted 1,470 data points covering 14 organolithium intermediates, each measured at 2–6 temperature levels with 5–8 residence times per level. Residence times were manually corrected from heatmap axis annotations (logarithmic scales), and yields were independently verified against a YOLO-based OCR pipeline (99.6% agreement, 734/737 exact matches).

For each (intermediate, temperature) pair, the data were fitted to a competing first-order kinetics model:

$$\text{yield}(t_R) = y_{\max}\left(1 - e^{-k_f t_R}\right) e^{-k_d t_R} \tag{1}$$

where k_f and k_d are the formation and decomposition rate constants, respectively, and $t_{1/2} = \ln 2/k_d$. A total of 76 (intermediate, T) groups were fitted, of which 53 exhibited measurable decay.

Arrhenius analysis

For each intermediate, the temperature dependence of k_d was modeled by the Arrhenius equation:

$$\ln(k_d) = \ln(A) - \frac{E_a}{RT} \tag{2}$$

Activation energies ranged from 4.8 kJ/mol (m-lithiobenzonitrile) to 65.4 kJ/mol (o-bromoaryllithium), and half-lives at -40 °C spanned nearly four orders of magnitude: from 3.3 ms (o-iodoaryllithium) to 51.9 s (*tert*-butyl o-(lithio)benzoate). Arrhenius fits yielded $R^2 > 0.95$ for 10 of 14 intermediates.

Three mechanistically distinct classes emerged from the E_a landscape: (i) conjugation-stabilized ArLi (E_a = 4.8–41.3 kJ/mol, 8 species), where the electron-withdrawing group delocalizes the carbanion; (ii) α-elimination carbenoids (E_a = 16.2–49.2 kJ/mol, 3 species); (iii) benzyne-elimination ArLi (E_a = 59.7–65.4 kJ/mol, 2 species), proceeding through concerted 1,2-elimination.

Hammett correlation for meta/para-substituted ArLi

For the subset of conjugation-stabilized ArLi bearing *meta* or *para* substituents, the decomposition activation energy correlated strongly with the Hammett substituent constant:

$$E_a = -48.5\sigma + 36.1 \quad (r = -0.979,\ p = 0.021,\ n = 4) \tag{3}$$

The negative slope indicates that stronger electron-withdrawing groups (higher σ) lower the activation barrier. Paradoxically, these intermediates are *more* stable because the pre-exponential factor $\ln A$ decreases even faster—a manifestation of enthalpy–entropy compensation ($\ln A = 0.63E_a - 4.6$, $r = 0.987$). The isokinetic temperature ($T_{\text{iso}} = 191$ K ≈ -82 °C) is close to the standard organolithium working temperature (-78 °C, dry ice/acetone), explaining why cryogenic conditions effectively equalize the stability of structurally diverse intermediates.

Taft steric correlation for ortho-substituted ArLi

The four *ortho*-alkoxycarbonyl ArLi intermediates (R = Me, Et, iPr, tBu) share identical Hammett σ values (0.45) but differ dramatically in stability. Their half-lives correlate with the Taft steric parameter E_s:

$$\log(t_{1/2}) = -1.13\,E_s + 0.04 \quad (r = -0.969,\ p = 0.031,\ n = 4) \tag{4}$$

The *tert*-butyl ester ($E_s = -1.54$) is 86× more stable than the methyl ester ($E_s = 0.00$) at −40 °C ($t_{1/2}$ = 51.9 s vs 602 ms). Notably, the same $CO_2{}^t$Bu group at the *ortho* position is 10.8× more stable than at *para* ($t_{1/2}$ = 51.9 s vs 4.8 s), demonstrating that steric shielding of the C–Li bond outweighs electronic stabilization for *ortho* substituents.

Unified four-parameter model

The separate Hammett and Taft correlations reveal that *meta*/*para* and *ortho* ArLi follow distinct stability trends. The key difference is that *ortho* substituents engage in intramolecular Li···O chelation, stabilizing the C–Li bond beyond what electronic effects alone predict: the same $CO_2{}^t$Bu group at *ortho* yields E_a = 26.8 kJ/mol versus 15.0 kJ/mol at *para*, an 11.8 kJ/mol stabilization from chelation. Meanwhile, the two *ortho*-halide ArLi (o-Br, o-I) decompose via concerted benzyne elimination with anomalously high E_a (59.7–65.4 kJ/mol) and $\ln A$ (∼36), a fundamentally different pathway from conjugation-stabilized species.

To capture electronic, steric, chelation, and benzyne-elimination effects in a single equation, we fitted a four-parameter model across all 10 ArLi intermediates with known Hammett σ:

$$E_a = -48.0\sigma + 8.0\,E_s + 25.5\,\delta_{\text{ortho}} + 44.4\,\delta_{\text{benzyne}} + 35.9 \tag{5}$$

$$(R^2 = 0.980,\ Q^2_{\text{LOO}} = 0.879,\ n = 10)$$

$$\ln A = -24.2\sigma + 6.7\,E_s + 15.6\,\delta_{\text{ortho}} + 29.7\,\delta_{\text{benzyne}} + 15.4 \tag{6}$$

$$(R^2 = 0.993,\ Q^2_{\text{LOO}} = 0.975,\ n = 10)$$

where $\delta_{\text{ortho}} = 1$ for *ortho*-ester ArLi (Li···O chelation) and $\delta_{\text{benzyne}} = 1$ for *ortho*-halide

ArLi (benzyne elimination); the two indicators are mutually exclusive. Half-lives at any temperature are computed via $t_{1/2}(T) = \ln 2/\exp[\ln A - E_a/(RT)]$, without recourse to enthalpy–entropy compensation relations.

Each coefficient has clear chemical meaning: -48.0σ captures electronic stabilization/destabilization of the carbanion; $+8.0\,E_s$ reflects steric shielding of the C–Li bond by the ester R group; $+25.5\,\delta_{\text{ortho}}$ quantifies the Li$\cdots$O chelation energy unique to *ortho* substituents bearing a coordinating group; $+44.4\,\delta_{\text{benzyne}}$ accounts for the high intrinsic barrier of concerted 1,2-elimination. Notably, E_a alone is not a reliable indicator of stability: benzyne-elimination species have the highest E_a (60–65 kJ/mol) yet the shortest $t_{1/2}$ (3–61 ms at −40 °C) because their $\ln A$ is commensurately high (∼36). Conversely, among conjugation-stabilized ArLi, stronger electron-withdrawing groups lower E_a but also lower $\ln A$—and $\ln A$ decreases faster, so the net effect is *greater* stability (e.g., *m*-CN-ArLi: $E_a = 4.8$ kJ/mol, $t_{1/2} = 11.9$ s vs PhLi: $E_a = 36.5$ kJ/mol, $t_{1/2} = 22.4$ s). The half-life is determined by the interplay of both parameters, which the unified model captures by treating E_a and $\ln A$ as independent response variables rather than linking them through an enthalpy–entropy compensation relation.

Validation by Arrhenius extrapolation to batch conditions

To test whether the extracted kinetic parameters are quantitatively predictive beyond the original flow conditions, we extrapolated the Arrhenius parameters of the four *ortho*-ester ArLi intermediates from the flow heatmap regime (−70 to 20 °C, 0.003–25 s) to batch conditions (−78 °C, 600 s). The predicted survival fractions were compared with independently measured batch yields from scope tables in the same publication (Table 3).

The *tert*-butyl ester prediction (58%) matches the experimental yield (61%) within 3 percentage points. For the ethyl and methyl esters, both prediction and experiment agree that the intermediates decompose completely under batch conditions. This close agreement—across a 100 °C temperature extrapolation and a 24× residence time extrapolation—validates

Table 3: Arrhenius extrapolation validation: predicted intermediate survival at −78 °C, 600 s versus measured batch yields.

Ester R	E_a **(kJ/mol)**	$t_{1/2}$ (−78 °C)	**Predicted survival**	**Batch yield**
*t*Bu	26.8	766 s	58%	61%
*i*Pr	38.0	257 s	20%	12%
Et	41.3	116 s	3%	0%
Me	36.5	24 s	0%	0%

the heatmap-derived Arrhenius parameters as quantitatively useful for reactor selection.

Predictive application

To close the prediction loop, two *para*-substituted aryllithiums not present in the training set were selected for experimental validation (Table 4). These two substrates bracket the Hammett σ range: p-CF_3 ($\sigma = +0.54$, strong EWG) and p-CH_3 ($\sigma = -0.17$, weak EDG). The predicted E_a and $\ln A$ values are obtained from the unified model (Equations 5–6 with $\delta_{\text{ortho}} = 0$, $\delta_{\text{benzyne}} = 0$, $E_s = 0$ for *para* substrates); $t_{1/2}$ values are computed via $t_{1/2} = \ln 2/\exp[\ln A - E_a/(RT)]$.

Table 4: Predicted kinetic parameters for two untested *para*-ArLi intermediates from the unified model (Equations 5–6).

Intermediate	σ	E_a^{pred}	$t_{1/2}$ (−40 °C)	**Survival (−78 °C, 600 s)**	**Regime**
p-CF_3-PhLi	+0.54	10.0 kJ/mol	11 s	≈ 0%	flow
p-CH_3-PhLi	−0.17	44.1 kJ/mol	17 s	75%	flow / batch

Both substrates have predicted half-lives of 11–17 s at −40 °C, placing them in the flow microreactor regime. However, their behavior diverges dramatically at cryogenic batch conditions (−78 °C, 600 s): p-CF_3-PhLi decomposes completely (survival ≈ 0%), whereas p-CH_3-PhLi survives (75%), comparable to PhLi. This divergence arises because E_a and $\ln A$ co-vary: p-CF_3-PhLi has low E_a (10.0 kJ/mol) *and* low $\ln A$ (2.4), so its rate constant is only weakly temperature-dependent and remains fast even at −78 °C; p-CH_3-PhLi has high E_a (44.1 kJ/mol) *and* high $\ln A$ (19.5), so cooling dramatically slows its decomposition. The unified model correctly captures this non-trivial compensation effect by modeling E_a and $\ln A$ separately, avoiding the spurious predictions that arise from enthalpy–entropy compensation shortcuts. These predictions are currently being validated experimentally using a T-mixer microreactor setup with MeOH quench (8 residence times × 4–5 temperatures per substrate).

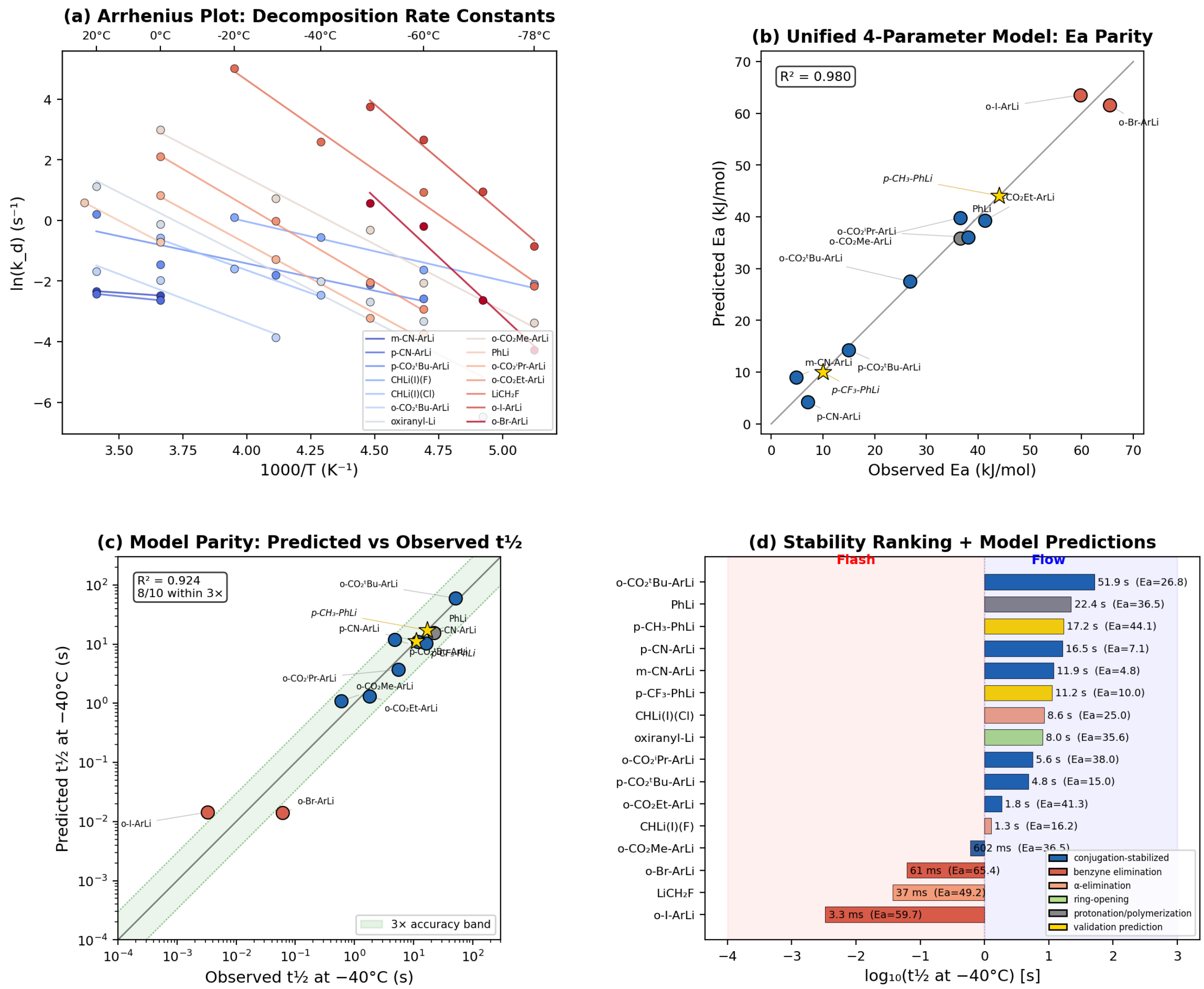


Figure 5: Kinetic analysis and structure–activity relationships for organolithium intermediates. (a) Arrhenius plots for 14 intermediates showing decomposition rate constants across temperature ranges. (b) Parity plot of observed vs. predicted E_a from the unified four-parameter model ($R^2 = 0.980$); gold stars mark predictions for two untested substrates. (c) Parity plot of observed vs. predicted $t_{1/2}$ at -40 °C (8/10 within 3× accuracy band); gold stars as in (b). (d) Stability ranking at -40 °C with Yoshida reactor regime zones (flash < 1 s, flow ≥ 1 s) and two model predictions (gold bars).

CONCLUSIONS

We have presented FlowFigTabMiner, the first fully automated multimodal pipeline for extracting structured process data from flow chemistry literature. On a 10-figure/10-table benchmark, the pipeline achieves an overall extraction F1 of 0.838, outperforming Gemini 3.1 Pro (0.833) and Claude Sonnet 4.6 (0.806), with a decisive advantage on quantitative figure extraction (F1 = 0.892 vs 0.575/0.561).

Applied to 125 organolithium flow chemistry publications, the pipeline constructed a 1,470-row kinetics dataset and a 1,267-row scope table dataset—the first with flow-condition coverage. From yield-vs-t_R heatmaps alone, we extracted decomposition activation energies for 14 organolithium intermediates spanning nearly four orders of magnitude in half-life (3.3 ms to 51.9 s at −40 °C).

A unified four-parameter LFER model emerged from the extracted data, integrating Hammett electronic (σ), Taft steric (E_s), Li$\cdots$O chelation (δ_{ortho}), and benzyne-elimination ($\delta_{\mathrm{benzyne}}$) effects into a single equation for all 10 ArLi intermediates with known σ ($R^2 = 0.980$ for E_a, $R^2 = 0.993$ for $\ln A$). By modeling E_a and $\ln A$ separately, the model captures the non-trivial compensation between activation energy and pre-exponential factor without reliance on enthalpy–entropy compensation relations. The +25.5 kJ/mol chelation term quantifies the stabilization conferred by intramolecular Li$\cdots$O coordination at *ortho* positions, while the +44.4 kJ/mol benzyne term accounts for the anomalously high barrier of concerted 1,2-elimination—an effect absent at *meta*/*para* positions and not captured by σ or E_s alone. Neither the individual Hammett/Taft correlations nor the unified model have been previously reported for organolithium decomposition kinetics. The kinetic parameters were validated by Arrhenius extrapolation to batch conditions (−78 °C, 600 s), where the predicted *tert*-butyl ester survival (58%) matched the experimental yield (61%) within 3 percentage points.

The unified model provides a practical tool for reactor selection: given the Hammett σ, Taft E_s, and substitution position, one can predict E_a, $\ln A$, and $t_{1/2}$ at any temperature. Two validation substrates (p-CF_3- and p-CH_3-bromobenzene) have been selected for experimental confirmation; both are predicted to require flow conditions at −40 °C ($t_{1/2}$ = 11–17 s), but they diverge at −78 °C (p-CF_3 decomposes completely; p-CH_3 survives at 75%, batch-compatible).

Limitations include the small number of intermediates (14, of which 10 are modelable ArLi with known σ) and limited data for benzyne-elimination species (2 substrates). Four non-ArLi intermediates (carbenoids, oxiranyl) decompose via fundamentally different mechanisms (α-elimination, ring-opening) and fall outside the model's scope. The pipeline architecture is domain-agnostic and can be extended to other organometallic systems (Grignard, organozinc) or domains where quantitative data is locked in scientific figures.

EXPERIMENTAL SECTION

Pipeline Implementation and Deployment

The pipeline was implemented in Python 3.10 with PyTorch 2.8.0 as the deep learning backend. The figure/table detection layer uses Florence-2-base[41] (TF-ID variant, HuggingFace `yifeihu/TF-ID-base`) for PDF page analysis. Object detection employs four YOLOv11 models (Ultralytics) at 1024 × 1024 px input resolution: three medium-backbone models (v11m) for figure and table segmentation, and one small-backbone model (v11s) for molecular structure detection. Table structure recognition uses the Microsoft Table Transformer v1.1 (TATR).[42] Text recognition employs PaddleOCR (PaddlePaddle 3.x, PP-OCRv5 server model) for tick labels and table cells, while molecular structure images are converted to SMILES via MolNexTR (1.06 GiB model).[43] LLM-based adjudication uses Qwen-3.5-Plus (temperature = 0.1) via cloud API to merge figure and table evidence into unified reaction records.

The system is deployed as four independent microservices on Google Cloud Platform Cloud Run: a TF-ID service (16 GiB memory, GPU-enabled for Florence-2 inference), a figure extraction service (8 GiB), a table extraction service (16 GiB, required for concurrent MolNexTR + TATR loading), and a frontend service. All model weights are stored in Google Cloud Storage and mounted via GCS FUSE at runtime. A web-based frontend provides drag-and-drop PDF upload with real-time progress tracking.

YOLO Model Training

Training data were prepared by manually annotating chemical figures and tables from flow chemistry publications using bounding boxes. Each annotation set was split into training and validation subsets (80/20). All models were trained at 1024×1024 px input resolution with the SGD optimizer (initial learning rate 0.01, momentum 0.937, weight decay 5×10^{-4}) and cosine learning-rate scheduling.

The v11m-fig-seg model (6 classes: caption, legend, subfigure marker, target image, x-axis title, y-axis title) was trained for 150 epochs with batch size 32. The v11m-fig-sca model (4 classes: data point, data value, x-tick label, y-tick label) was trained for 200 epochs with batch size 24 and Mixup augmentation ($\alpha = 0.15$) to improve robustness on overlapping detection targets. The v11m-tab-seg model (4 classes: table body, caption, footnote, reaction scheme) was trained for 200 epochs with batch size 32 and rectangular inference enabled to accommodate varying table aspect ratios. The v11s-tab-mol model (1 class: molecular structure) used the smaller YOLOv11s backbone and was trained for 200 epochs with batch size 32. The v11n-tab-scheme-seg model (4 classes: arrow, molecule, table-condition, table-mark) used the YOLOv11n (nano) backbone and was trained for 300 epochs with the AdamW optimizer (learning rate 5×10^{-4}, weight decay 0.05), full mosaic augmentation, and rectangular inference to handle the variable aspect ratios of reaction scheme diagrams.

Coordinate Mapping Algorithm

Coordinate mapping converts YOLO-detected pixel positions into physical units using the detected tick labels as calibration anchors. For each axis, the OCR-recognized tick labels are parsed to numerical values and paired with their pixel centroids. A RANSAC-based linear regression (pixel $\rightarrow$ physical value) is fitted to reject outlier tick detections, followed by monotonic sequence filtering (longest decreasing subsequence) to enforce axis ordering consistency. An alignment filter (bin size = 20 px) clusters tick labels by their transverse coordinate to discard off-axis spurious detections, and a ghost filter removes detections within 10 px of the axis lines (typically axis-label artifacts).

For logarithmic axes (common in residence-time plots), tick values spanning more than two orders of magnitude trigger automatic log-scale detection, and the pixel-to-value transform is fitted in log space. Dual y-axis figures are handled by partitioning y-tick labels into left and right groups based on their x-coordinate relative to the plot center. For heatmap figures, data-value labels (yield percentages) are detected directly and associated with the nearest (x, y) grid position. Legend matching assigns data series to conditions (temperature, substrate) via RGB color proximity between detected data points and legend swatches.

Evaluation Protocol

The evaluation set comprised 10 figures (F1–F10) and 10 tables (T1–T10) selected to cover the diversity of chart types (scatter plots, line plots, heatmaps) and table structures (simple, multi-header, with reaction schemes) encountered in flow chemistry publications. Ground-truth data were prepared independently by doctoral researchers through manual reading of each figure and table.

For figure extraction, each predicted data point (a numerical x–y pair) was matched to the nearest ground-truth point using greedy nearest-neighbor assignment. A match was counted as a true positive if the relative error was within 5% on both axes; unmatched predictions were false positives and unmatched ground-truth points were false negatives. For table extraction, exact string matching was applied to cell contents. SMILES correctness was evaluated by canonicalizing both predicted and ground-truth SMILES using RDKit; a match required identical canonical forms. The overall score was computed as the macro-average of figure F1, table F1, and SMILES F1.

VLM evaluation (Gemini 3.1 Pro and Claude Sonnet 4.6) was conducted on the first five figures and five tables (F1–F5, T1–T5) using identical prompts requesting structured JSON output.

Standard metrics were calculated based on True Positives (TP), False Positives (FP), and False Negatives (FN):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

$$mAP_{50} = \frac{1}{N} \sum_{i=1}^{N} AP_{50,i} \tag{10}$$

$$mAP_{50\text{–}95} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{K} \sum_{j=1}^{K} AP_{i,j} \tag{11}$$

Dataset Construction and Normalization

A corpus of 125 papers was selected by filtering for flow chemistry keywords ("organolithium", "aryllithium", "microreactor", "residence time", "Br/Li exchange", among 17 terms; full list in Supporting Information) applied to the first three pages of each PDF. Each paper was processed through the full pipeline, and the resulting reaction records were aggregated by LLM adjudication.

Post-processing normalization unified heterogeneous units reported across different publications: temperatures were converted to °C (from K or °F where necessary), residence times to seconds (from min, ms, or μs), and flow rates to mL min^{-1}. Solvent names were standardized using a vocabulary of 30+ aliases (e.g., "THF", "tetrahydrofuran", and "oxolane" mapped to a single canonical form). Reactor types were classified into flow, batch, and flash categories. SMILES enrichment was performed conservatively: compound names were matched against the Open Reaction Database (ORD) using exact string lookup, and only confirmed matches were assigned SMILES strings — no algorithmic name-to-structure conversion was employed. Final quality assurance was performed by doctoral researcher spot-checking of a random 10% sample.

Kinetic Modeling

Yield-vs-t_R data extracted from 20 papers (1,470 data points, 14 intermediates) were fitted using `scipy.optimize.curve_fit` to the competing kinetics model (Equation 1) for each (intermediate, temperature) pair, yielding the decomposition rate constant k_d and half-life $t_{1/2} = \ln 2/k_d$. A total of 76 fits were obtained (53 with measurable decay, 23 formation-only). For each intermediate, $\ln(k_d)$ values at 2–6 temperatures were fitted to the Arrhenius equation (Equation 2) via linear regression, giving the activation energy E_a and pre-exponential factor $\ln A$. Hammett σ constants were taken from Hansch, Leo, and Taft (1991); Taft E_s values from Taft (1956). Hammett and Taft correlations were fitted by ordinary least squares with p-values from two-sided t-tests. The unified four-parameter model (Equations 5–6) was fitted by OLS across all 10 ArLi with known σ, using `sklearn.linear_model.LinearRegression`; leave-one-out cross-validation (LOOCV) was performed to assess generalization (Q^2_{LOO}). The two indicator variables δ_{ortho} ($\text{Li}\cdots\text{O}$ chelation) and δ_{benzyne} (benzyne elimination) are mutually exclusive: *ortho*-ester substrates receive $\delta_{\text{ortho}} = 1$, $\delta_{\text{benzyne}} = 0$; *ortho*-halide substrates receive $\delta_{\text{ortho}} = 0$, $\delta_{\text{benzyne}} = 1$; all others receive both $= 0$. Arrhenius extrapolation to batch conditions used $k_d(T) = \exp[\ln A - E_a/(RT)]$ and survival fraction $= \exp(-k_d \cdot t)$.

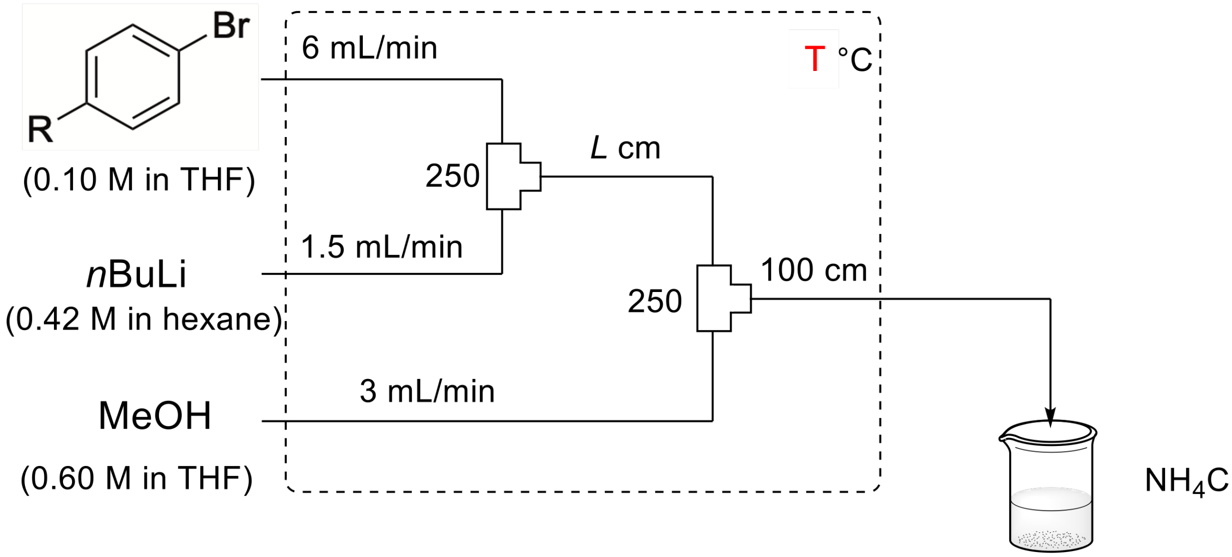


Figure 6: Validation experiment for predicted ArLi stability. R = CF_3, CH_3. (a) Reaction scheme: bromobenzene $\xrightarrow{n\text{-BuLi}}$ aryllithium $\xrightarrow{\text{MeOH}}$ quench. (b) Microreactor setup with T-mixer and MeOH quench. (c) Predicted vs measured survival fractions at −40 °C across 8 residence times.

Validation of Kinetic Predictions

ASSOCIATED CONTENT

Data Availability Statement

- The source code for the **FlowFigTabMiner** pipeline is publicly available at `https://github.com/wzjeh/FlowFigTabMiner`.
- The trained YOLO models are available on Hugging Face (wyzhaoc/YOLO11). The MolNexTR weights[43] are available at HuggingFace (CYF200127/MolNexTR). Third-party models (TF-ID, TATR, PaddleOCR) are downloaded automatically on first run.
- The organolithium kinetics dataset (1,470 rows), scope table dataset (1,267 rows), corpus DOI list, and dataset documentation are deposited on Zenodo: DOI: 10.5281/zenodo.19436601.
- A cloud-deployed web tool is accessible at FlowFigTabMiner.

Supporting Information Available

The Supporting Information is available free of charge at `https://pubs.acs.org/`.

- S1: Flow chemistry paper filtering keywords (15 keywords)
- S2: YOLO model training curves, per-class detection metrics, and confusion matrices for all five models
- S3: Tab-scheme-seg model training results, validation examples, and per-class performance
- S4: Complete LLM prompts used in the pipeline (local variable builder for figures and tables, global assembly adjudication prompt with 15 extraction rules)
- S5: VLM benchmark prompts for heatmap figure and table extraction
- S6: Three-way benchmark detailed results (per-task precision/recall/F1 and per-sample breakdown for F1–F10, T1–T10)
- S7: Complete list of 125 papers in the organolithium corpus

AUTHOR INFORMATION

Corresponding Author

Aiichiro Nagaki – Department of Chemistry, Faculty of Science, Hokkaido University, Sapporo 060-0810, Japan; orcid.org/0000-0002-8264-6205; Email: nagaki@sci.hokudai.ac.jp

Authors

Notes

The authors declare no competing financial interest.

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